

Course Overview

COMP3230B
Principles of Operating Systems

Course Info.

- **Subclass B** of Principles of Operating Systems
- Instructor
 - Prof. Zuming Jiang
 - Homepage: <https://jzuming.github.io/>
 - Email: jzuming@hku.hk
 - Office: Rm 301F, Chow Yei Ching Building
 - Office hours:
 - Monday: 9:00am – 10:00am (by appointment)

Research Experience in Operating Systems

- **TDSC 2023.** Testing Error Handling Code with Software Fault Injection and Error-Coverage-Guided Fuzzing.
- **NDSS 2022.** Context-Sensitive and Directional Concurrency Fuzzing for Data-Race Detection.
- **TSE 2022.** Hybrid Static-Dynamic Analysis of Data Races Caused by Inconsistent Locking Discipline in Device Drivers.
- **ISSRE 2019.** Fuzzing Error Handling Code in Device Drivers Based on Software Fault Injection.
- **SANER 2019.** Detecting Data Races Caused by Inconsistent Lock Protection in Device Drivers.

Talk to me if you have **new** ideas for OS research!

New:

Previous: study all related literatures, figure out the differences from existing approaches.

Now: ask your LLMs.

Tutors

- Mr. Yu, Miao
 - Email: miaoyu.hku@connect.hku.hk
 - Tutorial 2 + PA1 + PS2
- Mr. Pei, Yuxing
 - Email: yuxing.pei@connect.hku.hk
 - Tutorial 3 + PA2 + PS1
- Miss Gao, Mingyan
 - Email: mingyan.gao@connect.hku.hk
 - Tutorials 1 & 4 + PS3

PA: Programing Assignment

PS: Problem Set Assignment

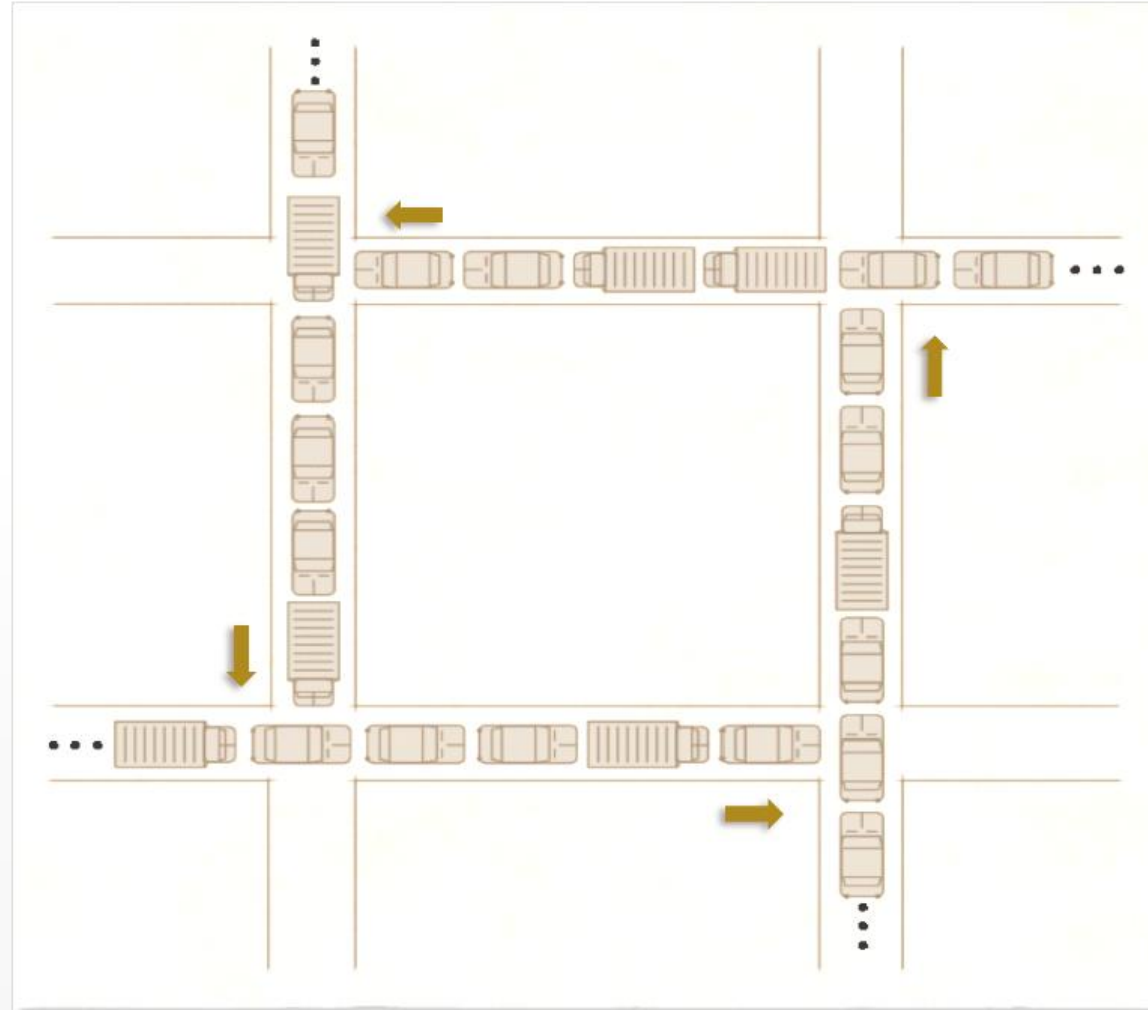
Moodle Course Site

- Homepage
 - Moodle course ID COMP3230_1B_2026
 - <https://moodle.hku.hk/course/view.php?id=139745>
- Course Information
- Teaching plan
- Reading List
- Lecture Notes
- Assignments' information and **submission of assignments**
- Announcements and Updates
- Web references
- Discussion Forum
 - Peer-to-peer discussions & asking questions

Course Objectives

- To present the fundamentals of operating systems, to study the **details of the operations and principles** behind the design of various components of an operating system
- Understand the **major components** that constitute the modern operating systems
- Learn the **underlying mechanisms and policies** of the operating systems and the implications of resulting **design choices**

Daily life problem



Daily life problem



Intended Learning Outcomes

- On successful completion of the course, students should be able to:
 - **[ILO1 – Fundamentals]** discuss the characteristics of **different structures** of the Operating Systems (such as microkernel, layered, virtualization, etc.) and identify the **core functions** of the Operating Systems.
 - **[ILO2 – Principles]** explain the **principles behind** the core functions and **compare the algorithms** on which the core functions of the Operating Systems are built on.
 - **ILO 2a** – managing **processes/threads** and sharing of **CPU resource**
 - **ILO 2b** – managing and allocating **memory resource** effectively
 - **ILO 2c** – how to support **concurrency and synchronization control** between processes/threads
 - **ILO 2d** – managing and allocating **persistent data storage**

Intended Learning Outcomes

- students should be able to:
 - **[ILO3 – Performance]** analyze and evaluate the algorithms of the core functions of the Operating Systems and explain the major performance issues with regard to the core functions.
 - **[ILO4 – Practicability]** demonstrate knowledge in applying system software and tools available in modern operating system (such as threads, system calls, semaphores, etc.) for software development.

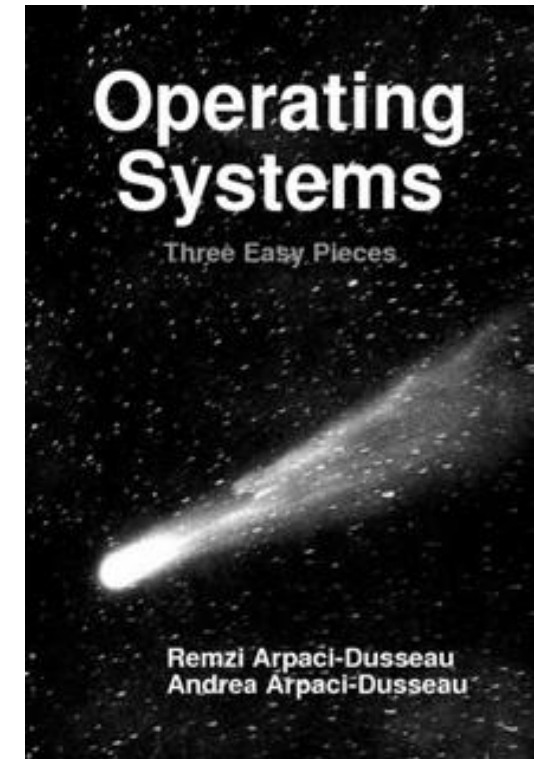
References

- **Required online textbook**

- Operating Systems: Three Easy Pieces
 - by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau
- <http://pages.cs.wisc.edu/~remzi/OSTEP/>

- Other references

- Deitel et. al, Operating Systems, 3rd Edition, Prentice Hall
- Silberschatz et. al, Operating System Concepts, 10th Edition, John Wiley
- William Stallings, Operating Systems: Internals and Design Principles, 9th Edition, Prentice Hall
- Bovet et. al, Understanding the Linux Kernel, 3rd Edition, O'Reilly
- Russinovich et. al, Windows Internals, 7th Edition, Microsoft Press
- Other online references



Assessment Tasks

- Continuous Assessment - total 50%
 - 4 take-home lab exercises – 4%
 - 3 problem-set assignments – 18%
 - Two programming assignments – 28%
 - No midterm quiz
- 3 hours final examination – 50%
 - **Open-book examination**

Slip Days (Late Submission) Policies

- <https://moodle.hku.hk/course/section.php?id=1603507>
- Only apply to problem-set and programming assignments
- Each student has five slip days
 - A student can use one slip day for pushing the assignment deadline by one day
- Assignment deadlines can be pushed back by at most 3 days
- After a student used up all slip-day credits, assignments that are submitted late will be graded but with a heavy penalty of 20% per uncovered day

Teaching Plan & Schedule

Week	Monday (CPD-3.04)	Friday (CPD-3.04)	Activities
Sep 4		- Course Overview - Introduction to OS	Release mini-labs
Sep 7 - 11	- Introduction to OS - Process abstraction	Process abstraction	
Sep 14 - 18	Virtualizing CPU	Tutorial 1 - Warm up!	Release Problem-Set Assignment 1
Sep 21 - 25	Process scheduling	Tutorial 2 - Process - Programming Assignment 1 briefing	Release Programming Assignment 1
Sep 28 - Oct 2	- Thread abstraction - Concurrency	- Locks - Condition Variables	Problem-Set Assignment 1 Due
Oct 5 - 9	- Semaphores - Deadlocks	Tutorial 3 - Pthread	Release Problem-Set Assignment 2
Reading Week (Oct 12 - 16)			Programming Assignment 1 Due
Oct 19 - 23	Holiday	Address space abstraction	Problem-Set Assignment 2 Due - Release Programming Assignment 2
Oct 26 - 30	Tutorial 4 - Semaphores - Programming Assignment 2 Briefing	Address translation	
Nov 2 - 6	Segmentation	- Paging - TLB	
Nov 9 - 13	Paging smaller tables	Beyond physical memory	Programming Assignment 2 Due - Release Problem-set Assignment 3
Nov 16 - 20	File and directories	File and directories	
Nov 23 - 27	File system implementation	Advanced topics	Problem-Set Assignment 3 Due
Nov 30	Course summary		mini-labs 3 Due

Two Programming Assignments

- Ass 1 – Implement a Job Submission program (15%)
 - To have firsthand practice in designing and developing a shell-like Job Submission program
 - Involves execution and management of multiple processes and collection of processes' execution statistics
- Ass 2 - Design a compute-intensive multithread program (13%)
 - To have hand-on practice in designing and developing multithread programs using Pthread library
 - Involves synchronization and coordination between threads

Plagiarism

- Plagiarism is a disciplinary offense. Any student who commits the offense is liable to disciplinary action
- We will make use of software tools to check against your submitted assignments
 - If we identify any suspicious cases, we will invite you to explain to us
- We will follow the departmental guidelines on handling any cases relating to the practice of plagiarism by students

Achieve the Learning Outcomes

- Most topics involve discussion of **mechanisms and policies**. The materials are **quite technical** and include low-level details, which requires dedication of time to comprehend
- You are provided with **a reading list**.
 - To get the best learning experience and result, have a preview of those chapters **before attending** the class
- Designate 6 to 7 hours every week for reading the given lecture notes and readings
- Learn by exploring more about the OS systems

Achieve the Learning Outcomes

- **Actively participating** during lectures
 - Raise questions if you think that helps you to understand better
 - Or ask me after the lecture if appropriate
- In the assignments and examination, questions **won't always map directly** to the examples you have seen before as one of the goals of learning is to be able to **use the concepts and principles in new/unseen situations**
- Always work on the assignments **ASAP**
 - Give yourself enough time to plan for the work
 - Don't waste your time “spinning your wheels” needlessly; raise your questions if needed
 - Effectively utilize the discussion forum as a means of information sharing

Computing Platform

- Ubuntu 22.04 / 24.04
- Department's Linux Servers
- Windows Subsystem for Linux version 2 (WSL 2)
- Course's docker Ubuntu image (Windows & Apple)
- For details - <https://moodle.hku.hk/course/section.php?id=1689298>

Get the CS account

- For new CS, non-CS, and exchange students:
 - Get your CS account from
 - <https://intranet.cs.hku.hk/csintranet/newstudent.jsp>